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Machine-learning-driven discovery of metal–organic framework adsorbents for hexavalent chromium removal from aqueous environments

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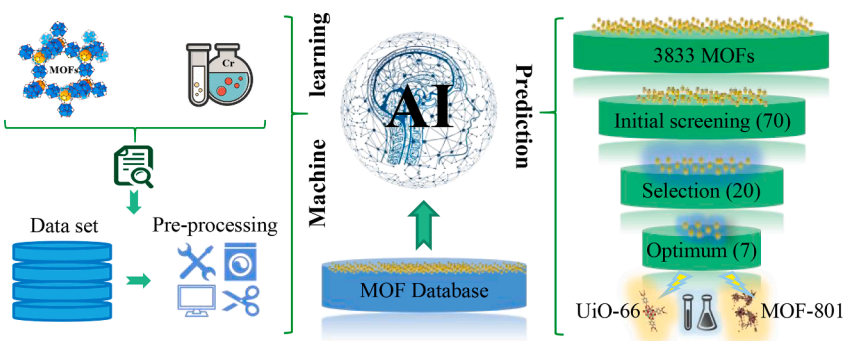
GRAPHICAL ABSTRACT



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ABSTRACT

Hypothesis: Metal-organic frameworks (MOFs) have been widely studied for Cr(VI) adsorption in water. Theoretically, numerous MOFs can be synthesised by assembling diverse metals and ligands. However, the traditional manual experimentation for screening high-performance MOFs is resource-intensive and inefficient.
Experiments: A screening strategy for MOFs based on machine learning was proposed for the adsorption and removal of Cr(VI) from water. By collecting the characteristics of MOFs and the experimental parameters of Cr (VI) adsorption from the literature, a dataset was constructed to predict the adsorption performance. Among the six regression models, the model trained by the extreme gradient boosted tree algorithm had the best performance and was used to simulate the adsorption and screen potential high-performance adsorbents.
Findings: Structure-property analysis indicated that prepared MOF adsorbents with properties of 0.37 < largest cavity diameter < 0.71 nm, 0.18 < pore volume < 0.57 cm^3/g, 412 < specific surface area < 1588 m^2/g, 0.43 < void fraction < 0.62 will achieve enhanced adsorption of Cr(VI) in water. High-performance adsorbents were successfully screened using a combination of machine-learning prediction and analysis. Experiments were conducted to verify the exceptional adsorption capacity of UiO-66 and MOF-801. This method effectively identified

Abbreviations: MOFs, Metal-Organic Frameworks; XGB, Extreme Gradient Boosted Tree; LCD, Largest Cavity Diameter; VF, Void Fraction; MT, Coordination Metal; ModM, Modification Methods; SA, Specific Surface Area; PD, Pore Diameter; PV, Pore Volume; CI, Initial Concentration; AD, Adsorbent Dose; Time, Reaction Time; Tem, Reaction Temperature.

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adsorbents and accelerated the development of new MOF adsorbents for contaminant removal, providing a novel approach for the discovery of superior adsorbents.

1. Introduction

Environmental pollution problems caused by the high toxicity of heavy metal ions in aqueous environments have drawn widespread attention [1,2]. Cr(VI) has strong carcinogenic properties and intense toxicity; it is pervasive in the environment and easily migrates and accumulates in living organisms. Therefore, effective removal technologies are highly sought after [3,4]. Currently, there are several approaches for the removal of Cr(VI) from aqueous media, including ion exchange, biological treatment, chemical reduction, precipitation, electrolysis and adsorption [5]. Adsorption is an effective, simple, and environmentally friendly method. Activated carbon, chitosan, and other resins are widely used for this purpose. However, activated carbon has drawbacks such as poor reproducibility, high processing costs, and limited application range [6]. Chitosan, on the other hand, is unstable and easily dissolves in acidic water [7,8]. Thus, the development of highly efficient adsorbents is particularly important in adsorption technology.

Metal-organic frameworks (MOFs) are periodic reticulated structures composed of coordination metals and organic ligands that have potential as adsorbents because of their tunable structures, large specific surface areas (SA), high porosities, and controllable functions [9]. Over the past decade, tens of thousands of MOFs have been synthesised and applied to gas adsorption and separation [10]. Studies on the adsorption of toxic pollutants such as antibiotics and heavy metals in water using MOFs have also been conducted but are in the early stages [1,11–13]. Theoretically, an infinite number of MOFs can be synthesised by assembling different metals and organic ligands. However, relying on extensive manual experiments to explore high-performance MOFs is a resource-intensive and inefficient trial-and-error process. Thus, a screening strategy to rapidly predict the adsorption performance of MOFs could have applications for screening superior adsorbents based on large databases of MOFs [14–16].

Data-driven machine learning (ML) is less dependent on prior knowledge than traditional programming methods and has the advantages of a low computational cost and powerful predictive capability, demonstrating a promising future for solving complex environmental data issues [17]. In recent years, ML has shown significant potential for screening functional materials [18–23], predicting gas adsorption [24], exploring adsorption mechanisms, and simulating pollutant removal [25–30]. Nguyen et al. developed a ML model for predicting the equilibrium adsorption capacity of biochar for drugs and explored the adsorption mechanism using literature data [25]. Molecular simulations combined with ML have become the new paradigm for discovering high-performance gas adsorbents through porous material screening. Recently, Wang et al. used an ML algorithm to explore the pattern of MOFs capturing chemical warfare agents from the air to assist in the screening of excellent MOFs [31]. However, the application of ML in adsorption is still in its infancy. In particular, a research gap remains in adsorbent screening for the removal of pollutants from water. It is difficult to obtain high-quality experimental data for ML modelling because of the scarcity of a direct database of sorption experiments that can be used for ML, which is the greatest obstacle to its development.

In this study, an adsorbent screening strategy was proposed for the adsorptive removal of Cr(VI) from water. A purely data-driven ML approach was used to predict the adsorption performance using the characteristics of the MOFs and the experimental parameters of Cr(VI) adsorption as data sources. Six regression models were trained and evaluated, including random forest (RF) [32], gradient boosting decision tree (GBDT) [33], extreme gradient boosting tree (XGB) [34], support vector regression (SVR) [35], linear regression (LR), and K-nearest neighbour (KNN) [36,37]. The optimal model was used to simulate the adsorption of Cr(VI) by the MOFs based on valid experimental parameters and the adsorption capacity was output. Potential high-performance adsorbents were screened using a combination of ML prediction and analysis. The feature importance and partial dependence plots (PDP) of the ML method were used to explain how the features affect the adsorption capacity and the degree of influence. The complete workflow for screening the MOF adsorbents is shown in Fig. 1. These results can accelerate the development of novel MOF adsorbents and serve as a reference for the rapid screening of high-performance MOFs for various operating scenarios in the future.

2. Data and methods

2.1. Data collection and pre-processing

The data used in the ML process were mainly from peer-reviewed publications (2014–2021, Elsevier, RSC – Royal Society of Chemistry, ACS publications) and 801 data points covering Cr(VI) onto 43 MOFs were extracted. During the process of literature selection, we excluded studies with unknown sources or significant missing data. With the adsorption capacity (Q; mg/g) as the prediction target, the collected data were broadly classified into two categories: adsorbent properties, including coordination metal (MT), modification method (ModM), SA (m^2/g), pore diameter (PD; nm), and pore volume (PV; cm^3/g), and experimental parameters, including initial concentration (CI; mg/L), adsorbent dose (AD; mg/L), reaction time (Time; min), pH, and reaction temperature (Tem; K). Because all data were manually collected from independent research literature without uniform standards and reliability assessments, two assumptions were made during the process of establishing the dataset: (1) all experimental data obtained from the literature sources could be objectively accepted and (2) the impact of data errors caused by experimental processes and data collection in each study on ML outcomes could be neglected. All data points were collected directly from graphs or text in the literature. Representative samples were given priority and duplicate samples were retrieved and eliminated from the entire dataset. For samples with a high number of missing values, selective elimination was performed. The inconformity of aperture data across publications may be detrimental to ML modelling. The theoretical mean PD (MPD; nm) was uniformly calculated and used as a feature. This approach can be understood as feature augmentation from a ML perspective. In addition, four quantitative variables, ionic charge (IC), atomic radius (AR), polarisability (Pol), and electronegativity (Ele), were used instead of the qualitative variable MT in the final dataset to benefit ML modelling.

The dataset was organised into 801 rows and 12 columns. To mitigate the risk of data leakage during the ML process, the data were divided into 43 groups based on the MOFs and arranged in sequential order. The missing pH and Tem values were assumed to be 7 and 298 K, respectively. Partially missing values for the four variables were as follows: PD (16.2 %), PV (14.2 %), SA (6.7 %), and CI (0.3 %). The RF algorithm was used to interpolate the data and fill in the missing data. Dimensional differences in the quantitative variables may negatively affect the predictive performance of the model. To accommodate ML modelling, the quantitative variables were standardised before training. In addition, qualitative variables were labelled and encoded in advance.

2.2. Statistical distribution and feature analysis

2.2.1. Statistical distribution

In the ML application, 11 features were used for the model training:

MT, ModM, SA, MPD, PD, PV, CI, AD, Time, pH, and Tem. Adsorption capacity (Q) was the prediction target. The statistical distributions of the 11 features used for modelling the dataset are listed in Table S1 and Figs. S1 and S2.

2.2.2. Correlation analysis

The Pearson correlation coefficient matrix for the quantitative variables in the dataset is presented in Fig. S3. There was an extremely significant correlation between the SA of the adsorbent and the PV ($p < 0.01$). Considering that SA and PV are important structural parameters of MOF, no exclusion operation was performed on any feature during model training.

2.2.3. Feature importance

The importance of each feature was calculated and ranked using the RF algorithm to evaluate its contribution to the adsorbed capacity. This method quantifies the impact of each feature on the forecasted results as a percentage and assesses the magnitude of the overall performance loss of the model after removing that feature [38].

2.3. Modelling and evaluation

The dataset was divided into training and test sets and a prediction model was built using the RF algorithm. As shown in Fig. S4, the prediction performance of the model improves as the percentage of the training set increases from 20 % to 90 %. The learning ability of the model for the relationship between features and adsorption capacity was positively correlated with the predictive performance. Therefore, a training set that included 90 % of the data was built to train the model and tune its hyperparameters. A test set that included 10 % of the data was used to evaluate the model prediction performance. Different types of ML algorithms have fixed hyperparameters and tuning their important hyperparameters to the optimum is the key to improve the model prediction performance. In the hyperparameters tuning process, the optimal hyperparameter combination was determined using the grid search method and 10-fold cross-validation after the parameter space of the model was determined [39,40]. The specific hyperparameter tuning process is described in the Text S1. In this way, an optimized prediction model is obtained. The test set was used to predict the adsorption capacity. The coefficient of determination (R^2) was calculated to evaluate the model performance and the mean absolute error (MAE) and root mean square error (RMSE) were applied to assess the model prediction accuracy.

Six ML regression algorithms were utilised in this study to develop prediction models, including RF based on the bagging method, GBDT based on boosting, XGB, LR, KNN, and SVR with nonlinear kernel functions. After training, a comprehensive comparison of the performance metrics among the six models, particularly R^2, was conducted to select the high-performance models for simulating the adsorption experiments. All the data processing and ML methods described here were performed in the Python programming language (version 3.8) using the open-source scikit-learn library [41].

2.4. Simulation of Cr(VI) adsorption by MOFs

2.4.1. Determination of experimental parameters

To facilitate the screening of adsorbents for Cr(VI) in water through simulation of adsorption by MOFs, it is necessary to determine effective experimental parameters in advance. A PDP was used to visualise how the experimental parameters affected the adsorption capacity. The PDP retained one or two target features as well as other marginalised features. A sensitivity analysis of the target features was provided by PDP [42].

2.4.2. Establishment of a data set of MOFs

A commercially available experimental metal–organic framework

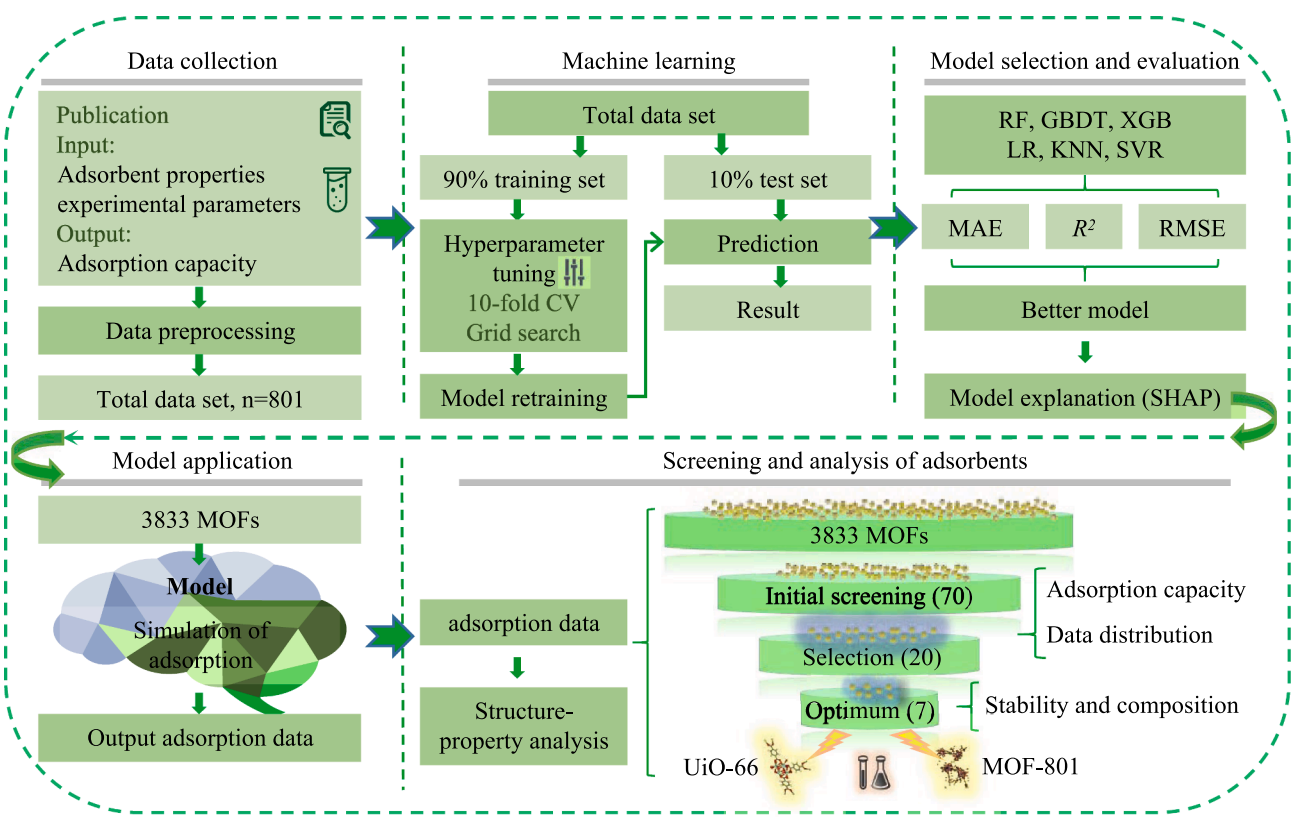

Fig. 1. Workflow for MOF adsorbent screening.

(CoRE MOF 2019) was used as the MOF data source. This comprehensive database contains 12,020 structurally diverse MOFs and provides a complete dataset of structural information that can be directly used for simulation calculations. Based on the CoRE MOF, high-throughput screening of a large number of MOF structures was implemented to find a subset of potentially promising structures and mine potential high-performance materials in detail [43]. Most metal MOFs, except for Zr-based MOFs, are generally unstable in aqueous solutions, especially in acidic (or basic) solutions. Therefore, only a few MOFs based on metal ions such as Zr, Cr, Zn, Fe, Cu, Co, and Ag have been applied to the study on the removal of Cr(VI) from water by adsorption [11]. Consequently, 3833 structural data points for MOF based on seven coordination metals were extracted from the CoRE MOF database. Among these were 103 Zr-based MOFs, 1581 Zn-based MOFs, 172 Fe-based MOFs, 1167 Cu-based MOFs, 564 Co-based MOFs, 229 Ag-based MOFs, and 17 Cr-based MOFs. Five features of the MOFs database in agreement with the training set were incorporated: MT, accessible SA, theoretical MPD, largest cavity diameter (LCD), and accessible PV. Considering the strong positive correlation between the LCD and dominant pore diameter in the MOF structure dataset established by Gülsoy et al., with a correlation coefficient of 0.99, the difference between the average values of the LCD and dominant pore diameter is less than 0.036 nm, with an error in a very small range [24]. Therefore, the LCD was used as the PD characteristic in the simulated adsorption experiments. Fig. 2(a) shows the structural relationships between SA, PD, and PV in the MOF dataset.

2.4.3. Formulating simulation strategy

Model fusion strategies have frequently been used to improve prediction accuracy because of the limitations of single models [44,45]. Moreover, owing to the limited data volume, the performance of the trained models may not be exceptionally robust. To further optimise the adsorbent screening, a multiple model combination strategy was used to simulate the adsorption experiments by connecting the first and second models in series for performance evaluation. First, under the determined experimental parameters, the best-performing model was employed to simulate the adsorption of Cr(VI) in water by the 3833 MOFs. The resulting adsorption capacity is referred to as the first adsorption dataset, from which a subset of MOFs with higher adsorption capacities was selected. Subsequently, this subset of MOFs was used as the input for the second-best-performing model to further simulate the adsorption experiments and the obtained adsorption capacity was designated as the second adsorption dataset.

2.5. Screening and analysis of adsorbents

2.5.1. Structure-property relationship of MOFs

The relationships between the adsorption data of the 3833 MOFs simulated by the first model and the geometric characteristics of the MOFs were visualised using univariate analysis. These relationships can provide valuable insights into the evolution of novel MOFs, facilitate the expedited experimental synthesis of new MOF adsorbents and serve as a crucial reference for large-scale screening of MOFs aimed at the removal of water pollutants [46].

2.5.2. Initial screening of adsorbents

As shown in Fig. 2(b), based on the first adsorption dataset, the

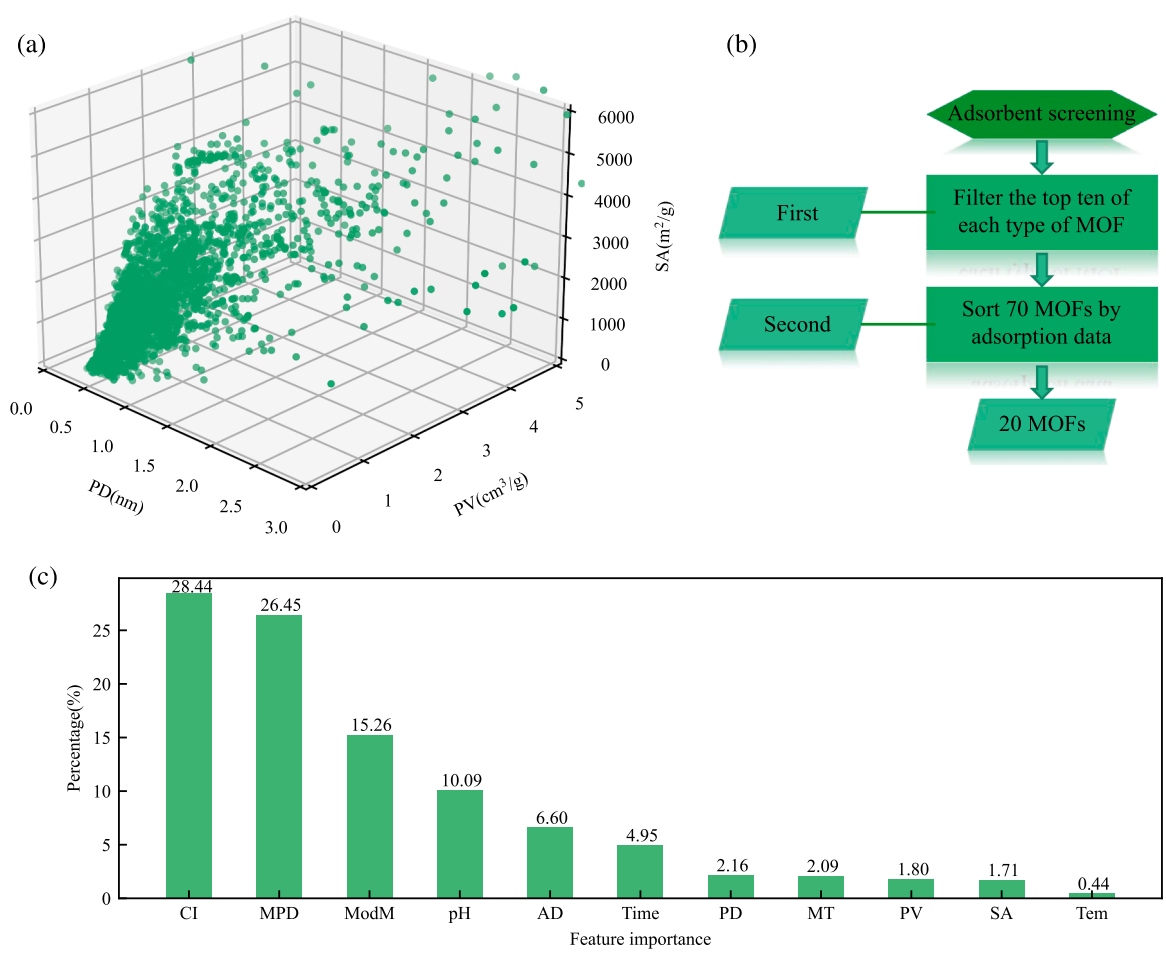

Fig. 2. (a) Relationship between structural data of MOFs; (b) adsorbent screening; and (c) feature importance.

predicted adsorption capacity was used as the ranking criterion. Considering the classification of the coordination metals, the top ten MOFs were selected for each type. Only one representative MOF was selected for MOFs within the same series with similar structures. Subsequently, considering the adsorption capacity and data distribution of the 70 MOFs toward Cr(VI) in water, based on the second adsorption dataset, the predicted adsorption capacity was ranked, and ultimately, 20 adsorbents were selected.

2.5.3. Adsorbent fine screen

However, in addition to possessing high adsorption capacity for Cr (VI) in water, they must also exhibit ideal stability as adsorbents. The Cambridge Crystallographic Data Centre (CCDC) provides comprehensive information on MOFs, including constituent ingredients, structural information, and relevant publications which facilitates further analysis of material properties.

For MOFs as adsorbent materials, high stability is essential to ensure the efficient removal of pollutants from water. Many researchers have proposed performance indicators for evaluating the stability of adsorbent materials, such as MOFs, including their stability after solvent removal, stability in aqueous solutions, and thermal stability [47,48]. The strength of the coordination bonds between the metal ions and organic ligands is a crucial factor affecting the stability of MOFs. According to the soft and hard acid-base theory, MOFs synthesised from hard acid metal ions and hard base ligands, as well as soft acid metal ions and soft base ligands, tend to exhibit higher stability in aqueous solutions [49]. Nandy et al. collected data from the literature to create a dataset on the stability of MOFs for ML modelling. They tested the performance of the model and accurately predicted the post-solvent removal stability and the thermal stability of MOFs, showing an accuracy of 78 % and MAE of 55 °C, respectively. The accuracy rate of MOFs classified as stable was even higher at 90 %. The reliability of the tool was independently validated using 10 MOFs, and the validation results are consistent with conclusions in the literature. They developed a predictive tool called MOFSimplify, which is available on a web interface and allows users to make stability predictions for new MOFs based on the stability data. We utilised this tool to predict the post-solvent removal stability and thermal stability of MOFs, aiming to offer a reference for the stability analysis of adsorbents [50].

3. Results and discussion

3.1. Feature importance

The contribution of each feature to adsorption capacity was determined using the RF algorithm. Fig. 2(c) illustrates that the CI of Cr(VI) in the solution exhibited a significant influence, contributing 28 % of the overall contribution. This finding is consistent with those of a previous study [25]. This is because the heightened gradients of Cr(VI) could surpass the mass transfer resistance from the aqueous phase to the solid adsorbent, thus promoting both the adsorption kinetics of the pollutants on the adsorbent surface and the subsequent pore diffusion at the later stage [51]. The feature importance ranking of pH was fourth, making it the second-most influential experimental parameter after CI in terms of adsorption capacity. This can be attributed to the alteration of the MOFs' charge characteristics caused by varying the pH values, which consequently affects the adsorption capacity for hexavalent chromium ions. In this study, it was crucial to establish effective experimental parameters before conducting the adsorption experiments. The ranking of importance of these parameters was as follows: CI > pH > AD > Time > Tem and the ranking of feature importance with respect to the adsorbent properties was MPD > ModM > PD > MT > PV > SA. MPD has the greatest influence on the adsorption capacity because the calculation of MPD is related to both the SA and PV of MOFs and, to some extent, encompasses the contribution of SA and PD.

Notably, ModM had a remarkable effect on the adsorption capacity, with a contribution of 15 %. The modification of the pristine MOFs effectively boosted the adsorption of Cr(VI) in water. Recently, researchers have introduced specific functional groups into MOFs to improve their adsorption of target pollutants. The dataset was divided into two parts: unmodified MOFs (416) and modified MOFs (385). The RF algorithm was employed to perform independent hyperparameter tuning on various datasets and train the models separately using the best hyperparameters to evaluate model performance. The results revealed that the MAE and RMSE of the former were 16.18 and 25.08, whereas those of the latter were 45.54 and 57.62, respectively. Regarding the modified MOFs, there were differences in the functional groups introduced, modification methods, and parameters across different studies. Thus, the performance of the final trained model was influenced to some extent. Therefore, the final data used for screening did not include modification parameters and relied solely on the structural characteristics of the MOF library to screen for superior MOF adsorbents.

3.2. Model selection

Based on the six ML algorithms, the model parameters were tuned to their optimal values and subsequently trained. Subsequently, the performances of each model were evaluated and compared. The optimal parameters of each model, MAE, RMSE, and R^2 predicted for the test dataset, are listed in Table S2. The fitting effects of the six models are shown in Fig. 3. Smaller MAE and RMSE values indicate higher model prediction accuracy and a larger R^2 value indicates better model fitting. By comparing the performance metrics of the six models, it can be concluded that tree-based ensemble learning algorithms, such as RF, GBDT, and XGB, can effectively capture the relationship between the features and adsorption capacity. However, only the RF and XGB models exhibited satisfactory performance, with R^2 values of 0.67 and 0.81, respectively. The XGB model exhibited the best performance, with an MAE of 13.66 and RMSE of 15.86. The optimal hyperparameter combinations were n_estimators = 250, max_depth = 15, min_child_weight = 7, gamma = 0.1, subsample = 0.8, colsample_bytree = 0.9, and learning_rate = 0.01. The RF and XGB models were explained using the Shapley Additive exPlanations tool (Figs. S5 and S6) [52,53]. The coefficients of determination of the other four models were either unsatisfactory or low. For example, the LR algorithm performed the worst, with an R^2 of −52.67, MAE of 205.2, and RMSE of 299.6. This was because LR is considered a relatively simplistic algorithm because of its sensitivity to outliers and inability to capture complex relationships in high-dimensional feature data [54].

In this study, the XGB model obtained from training was used as the first model for the simulated adsorption experiments and the RF model was used as the second model. Because the XGB model originates from the boosting concept, it can effectively reduce bias. The bagging-based RF model can effectively reduce variance. The XGB and RF models ranked as the top two in terms of performance. Combining them can complement each other and improve the excellence of the screened MOFs. For a detailed discussion of the specific functions of XGB and RF, please refer to Text S2 in the Supplementary Material.

3.3. Determining experimental parameters

As illustrated in Fig. 4, the combined influence of the CI of Cr(VI) and AD had a significant impact on the adsorption capacity. To fully explore the adsorption performance of MOFs when predicting the saturation adsorption capacity of MOFs for Cr(VI) in water, it is essential to increase the ratio of the Cr(VI) concentration to the adsorbent dosage, with pollutant concentrations exceeding 300 mg/L. Within 300 min, the reaction time had a significant effect on the adsorption capacity. As the reaction time increased, the adsorption capacity gradually decreased and reached a stable state. The adsorption equilibrium for most adsorption processes was reached within 12 h. The impact of pH on the adsorption capacity under acidic conditions was significant; however,

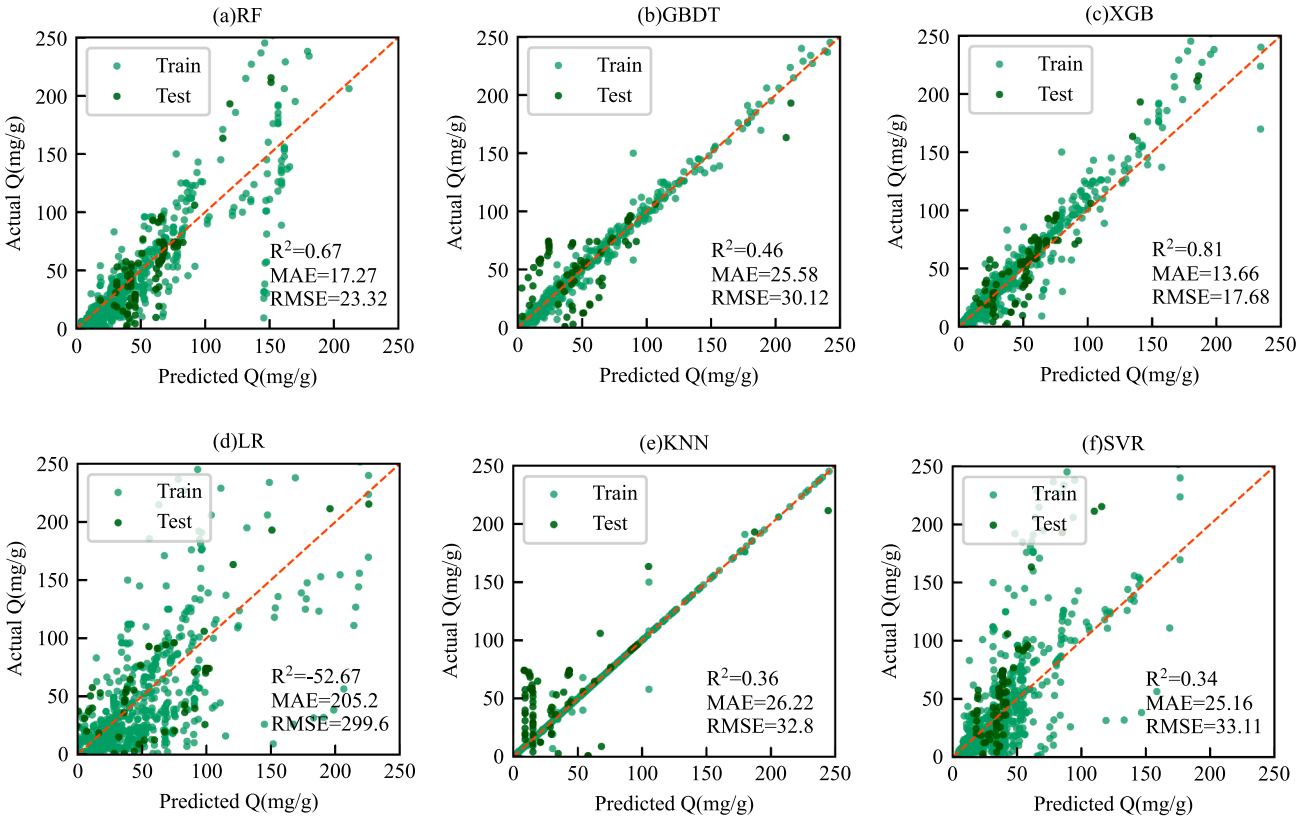

Fig. 3. Fitting effect diagram of six machine learning models.

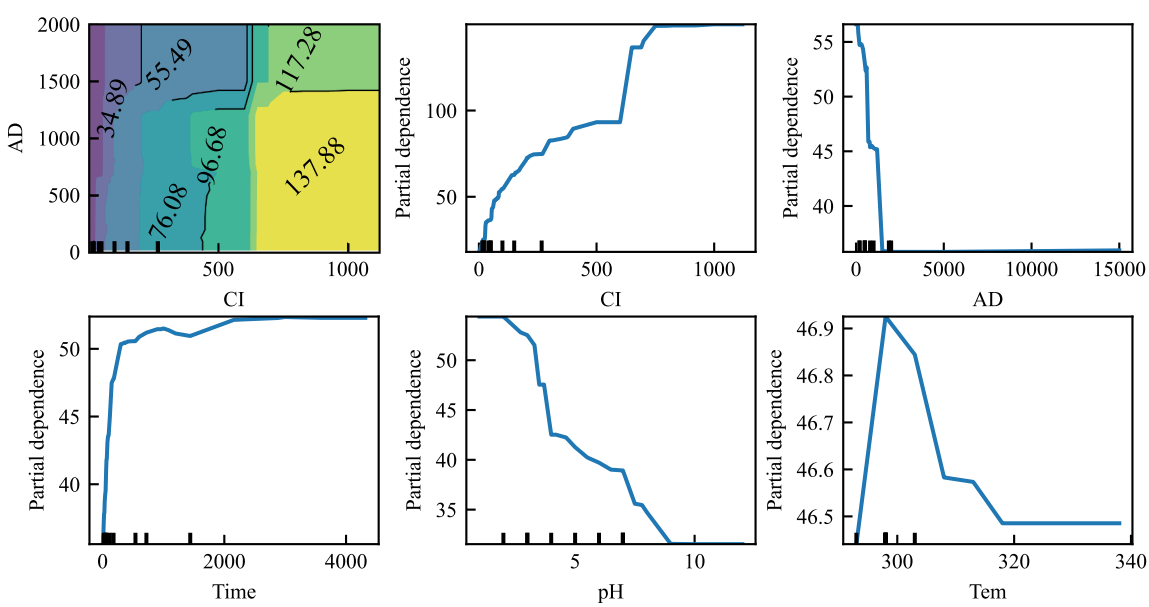

Fig. 4. PDP of experimental parameters, the blue line indicates the model's response to changes in the feature, and the black bars below each plot are used to display the density of the distribution of data values. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

many MOFs are unstable in acidic aqueous solutions. The effect of Tem on the adsorption capacity was not pronounced. Therefore, to support the adsorbent screening, a simulation experiment of MOFs adsorbing Cr (VI) in water was conducted under the following experimental parameters: CI (300 mg/L), AD (500 mg/L), Time (720 min), pH (7), and Tem (298 K).

3.4. Structure-Property relationship of MOFs

The correlation between the adsorption capacity predicted by the XGB model and the structural characteristics of the MOFs is shown in Fig. 5. The results show that the model effectively learned the relationship between the adsorption capacity and structural characteristics,

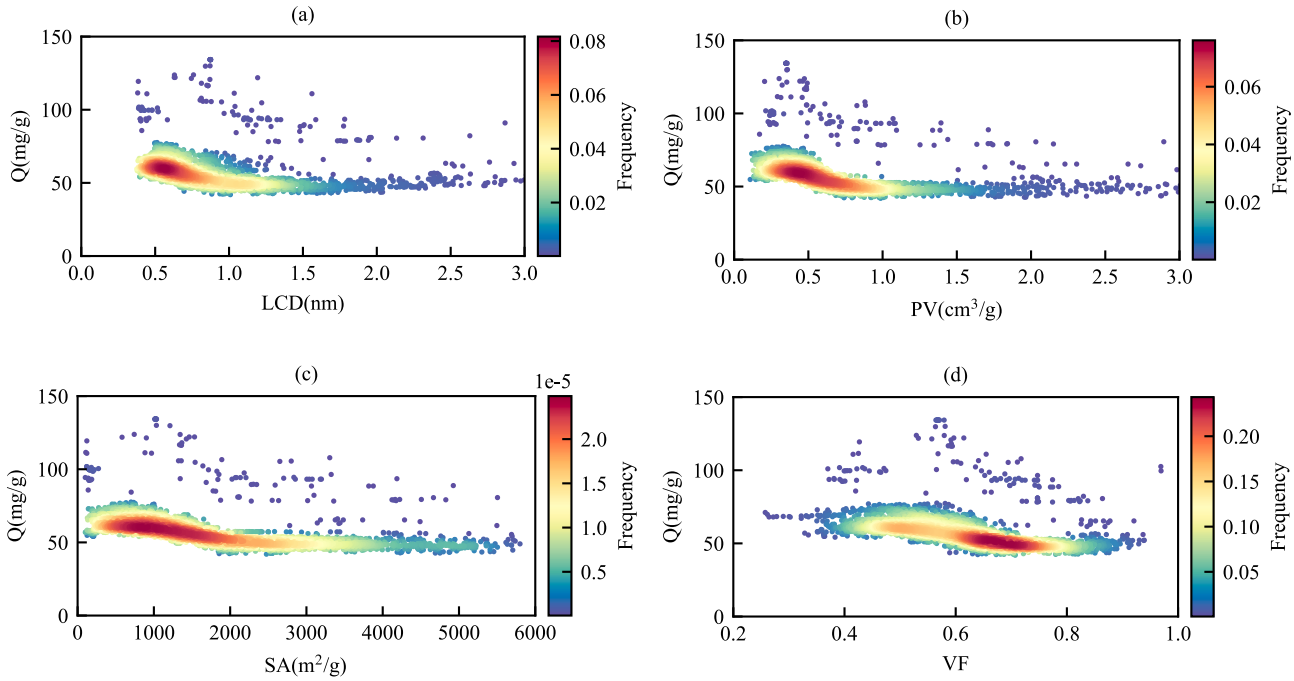

Fig. 5. Structure-adsorption capacity relationships of MOFs.

including the void fraction (VF). Consequently, the optimal structural range of MOFs for achieving a high adsorption capacity for Cr(VI) in water was identified. Thus, the preparation of MOF adsorbents with $0.37 < LCD < 0.71$ nm, $0.18 < PV < 0.57$ cm^3/g, $412 < SA < 1588$ m^2/g, and $0.43 < VF < 0.62$ will have enhanced Cr(VI) adsorption capacity in water. These findings will expedite the experimental design of novel MOF adsorbents. Dichromate ($Cr_2O_7^{2-}$) with a size of ca. 0.6×0.39 nm^2 is the main form of hexavalent chromium in water[55]. The Cr(VI) concentration is below the acceptable limit for the drinking water standard (100 μg/L by the U.S. Environmental Protection Agency).

3.5. Screening of adsorbents

According to the established screening procedure, a hierarchical screening of 3833 MOFs was conducted based on the predicted adsorption capacity. This study prioritised the predicted data from the XGB model to select the top 10 MOFs (out of 70) for each class of coordination metal MOFs from a total of 3833 MOFs. The 70 MOFs were then reranked based on the predicted data from the RF model to make further selections according to the classification of coordination metals. During the MOF selection process, preference was given to Zn- and Cu-based MOFs that were highly abundant in the dataset, as well as to Zr-based MOFs known for their superior stability. Ultimately, 20 MOFs stood out and are listed in Tables S3 and S4. From the model itself, the screened 20 MOFs possess high Cr(VI) adsorption capacity in water. However, further analysis and screening are necessary to assess the feasibility of MOFs as adsorbents for the removal of pollutants from water. For this purpose, information regarding the composition, structure, and relevant literature for MOFs were collected and their stability and thermal stability after solvent removal were predicted. To establish a reference for further screening, the structure-adsorption capacity relationships of the 20 MOFs were ranked. Because of the different coordination metal types, the coordination metal information was normalised. The purpose of this normalization was to shield the potential influence of coordination metals on the adsorption performance. By setting the coordination metal feature to a uniform value, any variation in its effect on the predicted adsorption capacity among different MOFs was eliminated. This approach enables a more accurate comparison and evaluation of other structural features related to adsorption capacity among the MOFs. The XGB model was used to predict the adsorption capacity under the determined experimental parameters and was ranked as an indicator for further screening. Information regarding the 20 MOFs is presented in Table S3.

Initially, three MOFs without stability data were excluded and are listed in Group D of Table S4. Subsequently, depending on the predicted stability values, MOFs with stability values <0.6 upon solvent removal and thermal stability <300 °C were excluded and placed in Group C. Lastly, referring to the ranking and stability assessment of MOFs, poor-performing MOFs were excluded and listed in Group B. Eventually, seven potential excellent MOF adsorbents were identified and listed in Group A. Information on these seven superior MOFs is shown in Table 1. Some are already well known, such as UiO-66, which will facilitate further validation.

3.6. Adsorption validation

Among the seven selected MOFs, MOF-801 and UiO-66 had been investigated for their ability to adsorb Cr(VI) from water. As shown in Table 2, compared to the unmodified MOFs, MOF-801 and UiO-66 exhibited superior adsorption performances. MOF-801, consisting of a $Zr_6O_4(OH)_4$ cluster and a short linker of fumaric acid, was synthesised by the solvothermal method and studied for its adsorption performance toward Cr(VI) from an aqueous solution. All the adsorption experiments were performed in batch mode. The adsorption capacity of Cr(VI) in water by MOF-801 can reach 156 mg/g, ranking second with an initial concentrate range is 5–1000 mg/L [4]. UiO-66, which contains hexanuclear zirconium clusters linked by terephthalates, was synthesised using a hydrothermal method. Hexavalent chromium adsorption was studied using UiO-66 and all experiments were performed under batch conditions. The adsorption capacity of UiO-66 was 86 mg/g at a CI of 50 mg/L, ranking fifth, lower than that of ZIF-8 (1050 m^2/g). One reason for this is that ZIF-8 has a higher SA than UiO-66 (800 m^2/g), thus providing more adsorption sites [56].

We synthesised MOF-801 and UiO-66 using the methods described in the original publications [56,57] and conducted further experiments to validate their Cr(VI) adsorption capacities in water[4]. The sources and

Table 1
Information on 7 excellent MOFs.

Ref code	Synonyms	Coordination metal	Stability in aqueous solutions and acidic conditions	Thermal stability℃ & Stability upon solvent removal (Predicted)	DOI_public
RUBTAK04	UiO-66	Zr	Highly stable	>400	https://doi.org/10.1002/anie.201505461
SURHEU	UiO-66	Zr	Highly stable	>400	https://doi.org/10.1002/anie.201505461
SURKUN	UiO-66	Zr	Highly stable	>400	https://doi.org/10.1002/anie.201505461
BIBBUL	–	Zn	–	352.8 & 0.69	https://doi.org/10.1016/j.ica.2013.03.015
XUFPIZ	–	Co	–	406.8 & 0.9	https://doi.org/10.1016/j.inoche.2014.10.019
BOHJUF	MOF-801	Zr	Stable	>400	https://doi.org/10.1021/ja500330a
SETQEP	MIL-53 (Fe)	Fe	Stable	324.5	https://doi.org/10.1021/cm301242d

Table 2
Comparison of the adsorption capacities of Cr(VI) in water by MOFs.

MOF	Adsorption capacity(mg/g)	Concentration (mg/L)	pH	Ref.
UiO-66	164	100	7	this work
MOF-801	156	1000	–	[4]
MOF-801	144	100	7	this work
ZIF-8	112	50	3	[56]
UiO-66	86	50	3	
FIR-53	74	312	–	[58]
MIL-100(Fe)	74	100	2	[59]
Dy-MOF	63	–	–	[60]
MOF-5	62	40	4	[61]
SLUG-21	60	–	–	[62]
Cu-BTC	58	40	7	[6]
MIL-100(Fe)	54	10	2	[63]
MOF-867	54	400	7	[64]
MIL-101(Cr)	40	100	6	[65]
ZIF-8	25	20	7	[66]
ZIF-67	15	30	7	[67]
NU-1000	49	50	7	[55]
UiO-66	30	50	7	
UiO-67	20	50	7	
MIL-100(Fe)	7	50	7	
HKUST-1	3	50	7	
ZIF-8	<1	50	7	

purities of all chemicals are listed in Table S5. Cr(VI) solutions (200 mL) with different concentrations were prepared in 250 mL conical flasks. Subsequently, 5 mg of MOF-801 and UiO-66 were added to each flask, which were then sealed with a cling film and placed in a constant-temperature shaker at 170 rpm. The CI of Cr(VI) in the solution ranged from 10 to 100 mg/L, with the pH set to natural, and the oscillation time set to 600 min at room temperature (298 K). As shown in Table 2 and Fig. 6, the adsorption capacities of MOF-801 and UiO-66 were 144 and 164 mg/g, respectively, ranking first and second, respectively, when the CI reached 100 mg/L. These results further confirmed the excellent adsorption performance of both MOF-801 and UiO-66.

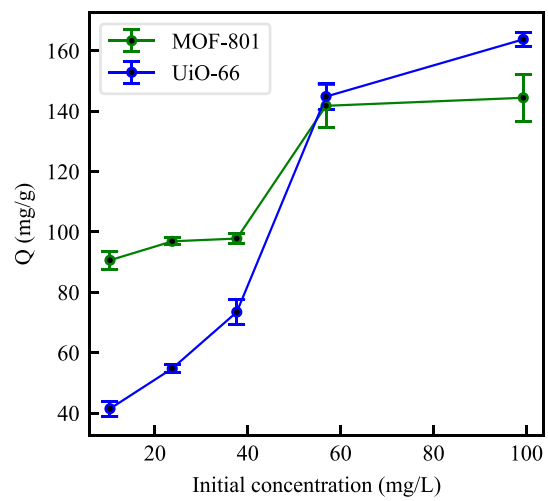

Fig. 6. Adsorption isotherms of Cr(VI) on MOF-801 and UiO-66 (Initial concentration = 10–100 mg/L, natural pH, adsorption time = 600 min, room temperature = 298 K, replication of adsorption experiments: n = 3).

4. Conclusions

Using a data-driven ML method, this study identified XGB as the prediction model with the best model performance. A high-precision prediction of the Cr(VI) adsorption performance of the MOFs was achieved and the range of structural characteristics of the high-performance MOF adsorbents was determined. Although some published studies have presented views on predicting the performance of MOF adsorbents, challenges in screening persist [37,68–70]. This study proposed a MOF adsorbent screening strategy by mining the adsorption rules and revealing the structure–activity relationship, which was successfully implemented to screen seven MOFs with the best adsorption properties from 3833 MOFs and verified the adsorption properties of two MOF adsorbents with high potential, UiO-66 and MOF-801, through experiments. This study provides new ideas for the optimisation and screening of adsorbents for the removal of Cr(VI) pollutants from water.

Nevertheless, there is still room for improvement in the performance of models used to simulate adsorption experiments. In addition to insufficient data, another significant reason for this is the specific adsorption caused by the presence of specific functional groups in certain MOFs, which is also an influential factor in the adsorption capacity. These factors limit the ability of the data to fully support model learning. Further, in-depth studies are required to address these limitations.

CRediT authorship contribution statement

Mingxing Jiang: Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Weiwei Fu:** Writing – review & editing. **Ying Wang:** Writing – review & editing, Data curation. **Duanping Xu:** Supervision, Funding acquisition. **Sitan Wang:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jcis.2024.02.084.

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